300–6000MHz Digital Noise Modulator

Controller App — User Guide

I. First-Time Connection Setup

Go to the **Settings** page and fill in the fields below.

Field	What It Is	Default
COM Port	Your USB↔RS422 adapter's port (check Windows Device Manager -> Ports if unsure)	—
Baud Rate	Communication speed	115200
Data Bits	Serial frame size	8
Parity	Error-checking mode	None
Module Address	The device's internal ID (0–199)	0
Auto Connect	Connect automatically on app startup	Off
Log Folder	Where the rotating file log is written	logs

Use **Query** / **Set** next to Module Address to read or change the device's internal address directly, instead of guessing.

Click **Save Configuration** to persist your settings — they are used the next time you connect or auto-connect.

II. Connecting

On the **Dashboard** page, pick your port (Refresh if it is not listed) and click **Connect**. The status label changes to “Connected” and the button becomes “Disconnect.”

If Auto Connect is enabled and a valid COM port is saved, this happens automatically at startup, before the window even appears.

III. Controlling the Device

Go to **Device Control**:

- **Output** — a toggle switch that flips the RF output on/off immediately.
- **Mode** — White Noise, Linear Sweep, Comb Spectrum, or Single. *Single is marked “(unconfirmed)” — its protocol byte value is a guess, not proven against real hardware. Selecting it and pressing Apply asks for confirmation first.*
- **Center Frequency** — 300–6000 MHz, with a step-size dropdown (1/10/50/100 MHz) and +/– buttons for quick adjustment.
- **Bandwidth** — 10 through 300 MHz. *300 MHz is marked “(unconfirmed)” for the same reason as Single mode.*

- **Power** — 0 dB (max), –6 dB, or –12 dB. (This is what the real vendor software calls “attenuation” — same setting, different name.)
- **Apply** — sends the above as one Signal Control command.
- **Read Device** — queries the device's actual current status and refreshes the Dashboard from the real response.

IV. Reading Status

The **Dashboard** shows live cards for Connection, Output State, Frequency, Bandwidth, Power, and Current Mode, plus a “Last Command” line — all update automatically whenever the device confirms a command or you read status.

If a command gets no response within 2 seconds, an amber warning banner appears on the Dashboard naming which command timed out — check wiring, module power, and module address, in that order.

V. Communication Log

The **Communication** page shows a running, timestamped log: incoming raw hex (RX), parsed frame descriptions, connection/disconnection events, errors, and timeouts. Use **Clear** to wipe it.

VI. First-Time Hardware Bench Test

Before trusting the full app against a device you have never connected before, proceed in this order:

- **Windows Device Manager** -> Ports (COM & LPT) — confirm the adapter shows up at all.
- Run the app and use **Read Device** on the Device Control page. Either the Dashboard populates with real values (success), or the amber warning banner appears within 2 seconds — check wiring, power, and module address, in that order.
- Once that works, use confirmed settings first (White Noise/Linear Sweep/Comb Spectrum, ≤250 MHz, default parity/baud/data bits) before trying anything flagged “(unconfirmed).”